

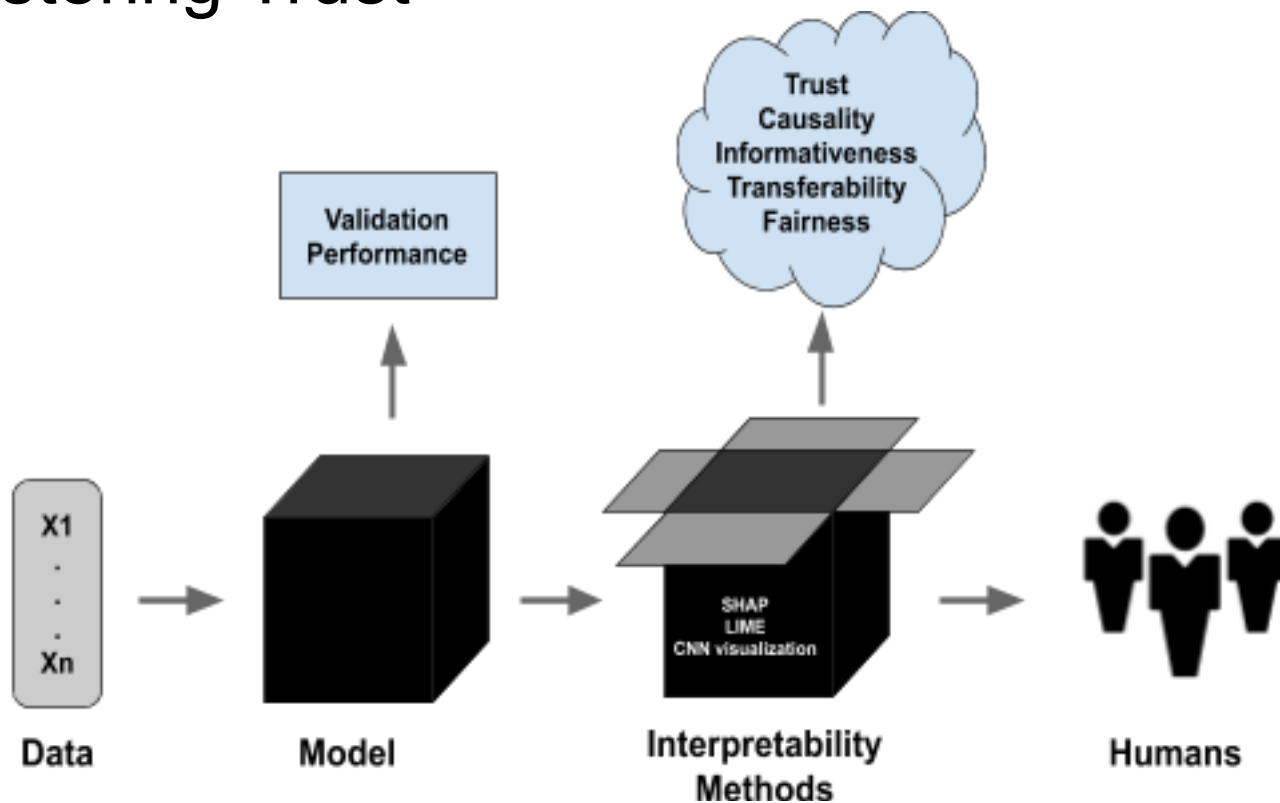
# Flag Representation Hypothesis

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Pedro Valois, Lincon Souza, Erica Shimomoto, Kazuhiro Fukui

# Interpretability

## ■ Fostering Trust



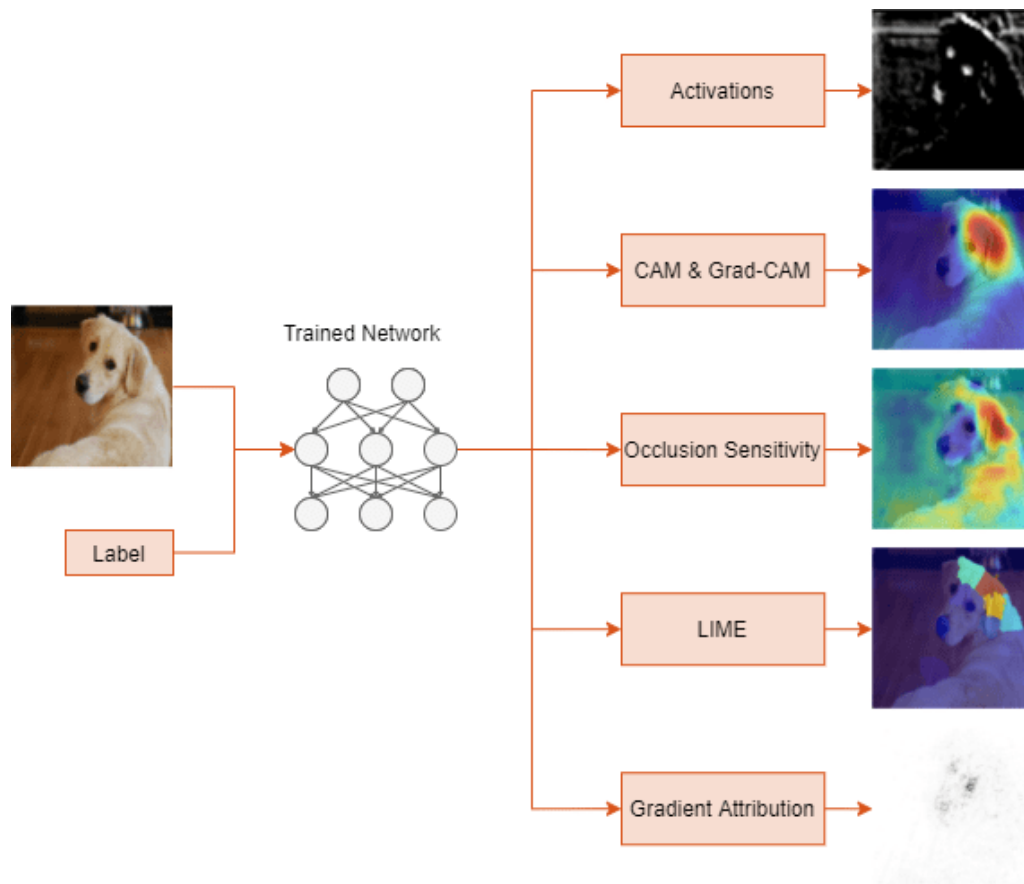
# Interpretability

- We need to fix when it breaks.



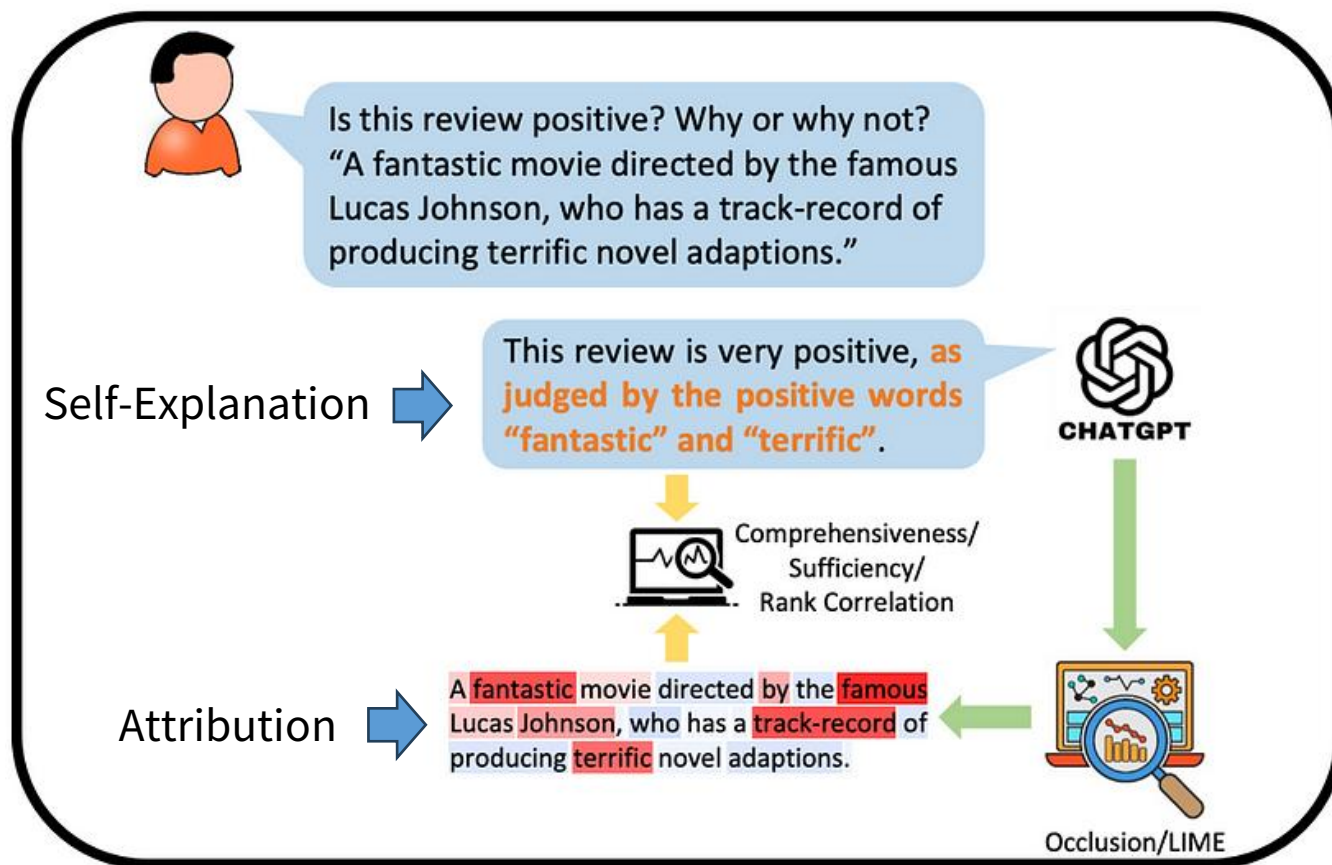
# Interpretability in Vision

## ■ Attribution: Where is the model looking at?



# Interpretability in Language

- Attribution is limited to classifications.

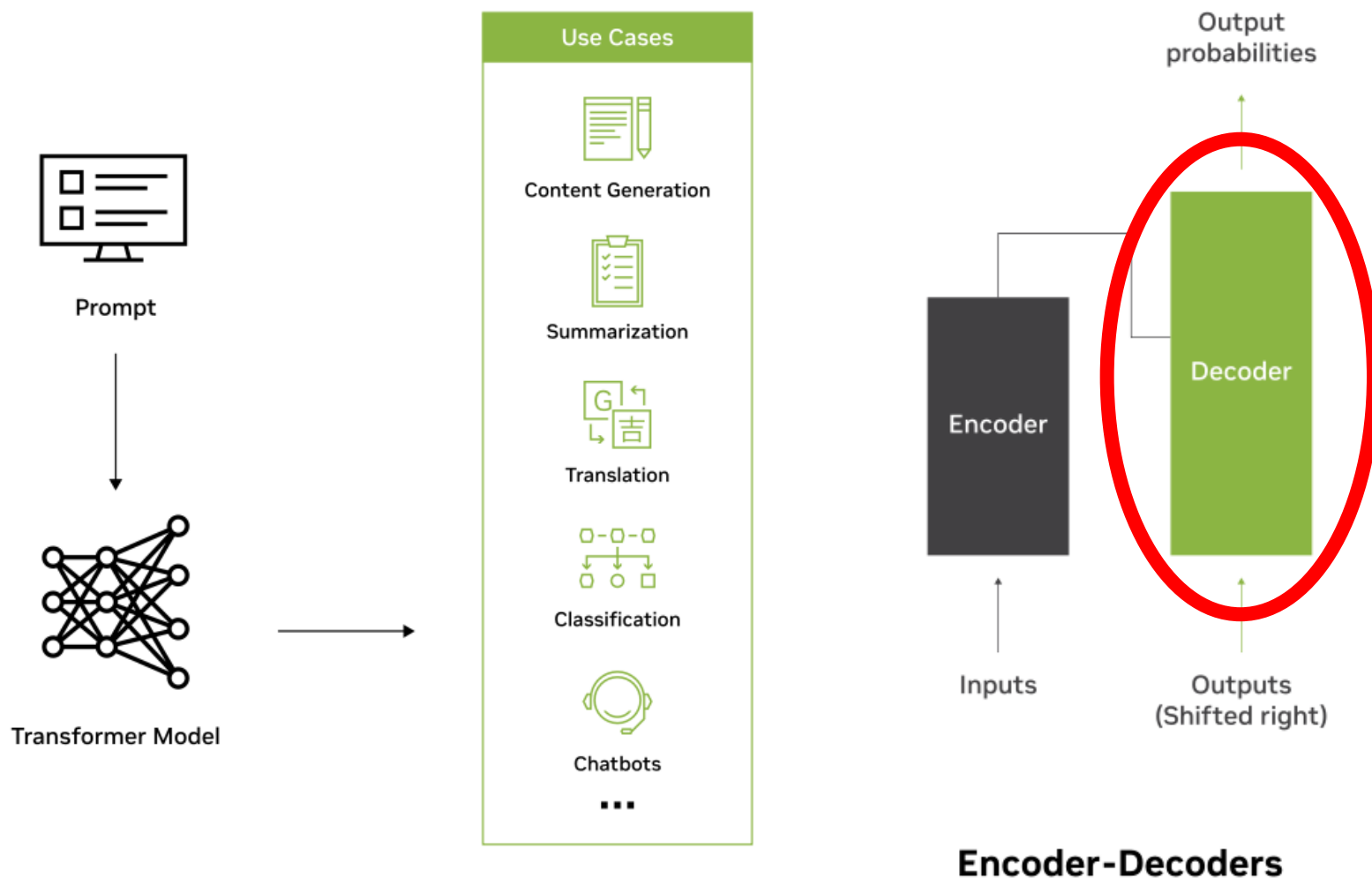


# What is this Research

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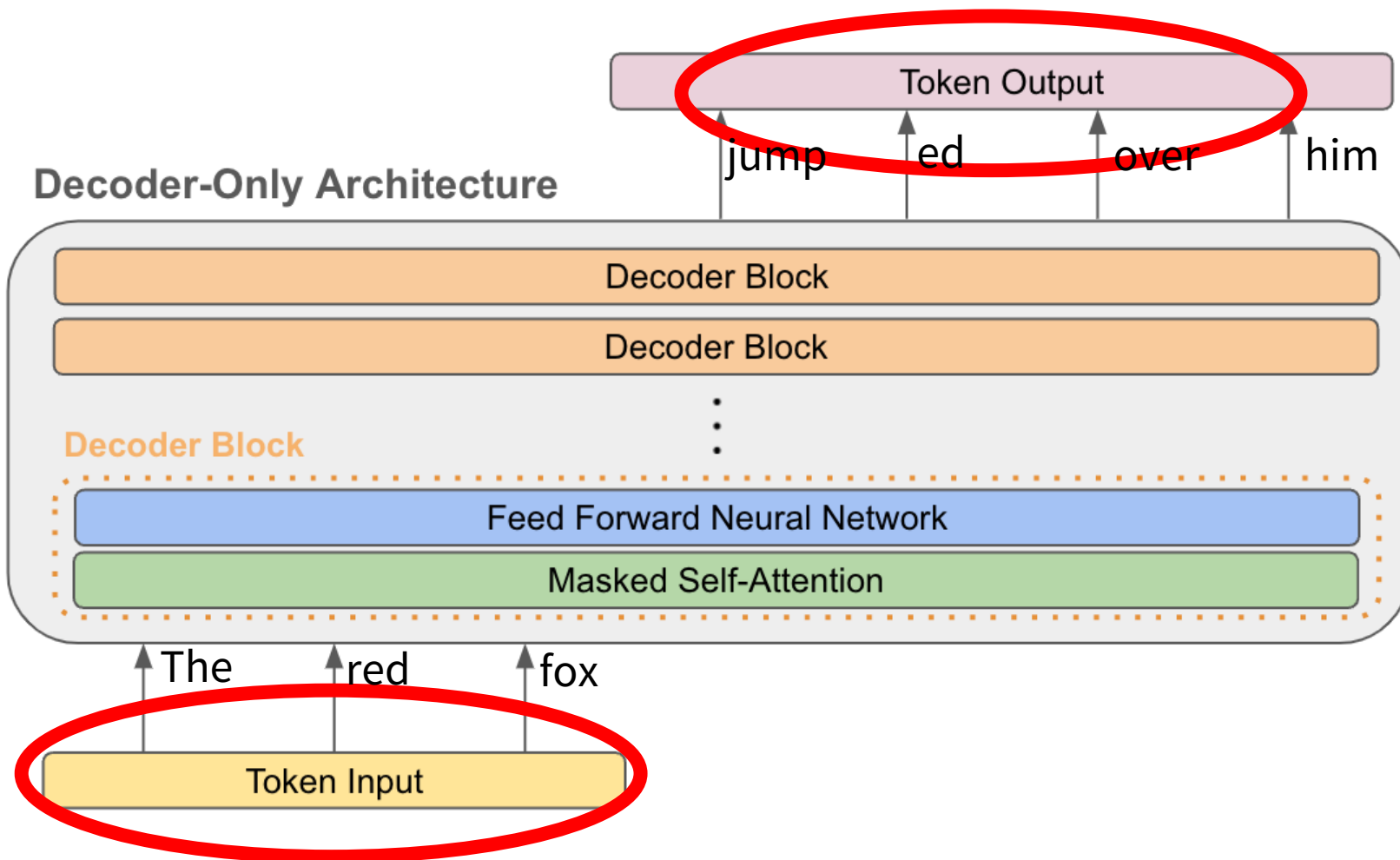
- LLM Interpretability is difficult.
- We model LLM features as Flag Subspaces.
- We are able to better understand and control LLMs with this proposal.

# What are Large Language Models?





# Decoder Only LLMs





# What is Tokenization?

## ■ How computers process text?

```
cook@pop-os:~$ ascii -d
```

|    |     |    |     |    |    |    |   |    |   |    |   |     |   |     |     |
|----|-----|----|-----|----|----|----|---|----|---|----|---|-----|---|-----|-----|
| 0  | NUL | 16 | DLE | 32 |    | 48 | 0 | 64 | @ | 80 | P | 96  | ` | 112 | p   |
| 1  | SOH | 17 | DC1 | 33 | !  | 49 | 1 | 65 | A | 81 | Q | 97  | a | 113 | q   |
| 2  | STX | 18 | DC2 | 34 | "  | 50 | 2 | 66 | B | 82 | R | 98  | b | 114 | r   |
| 3  | ETX | 19 | DC3 | 35 | #  | 51 | 3 | 67 | C | 83 | S | 99  | c | 115 | s   |
| 4  | EOT | 20 | DC4 | 36 | \$ | 52 | 4 | 68 | D | 84 | T | 100 | d | 116 | t   |
| 5  | ENQ | 21 | NAK | 37 | %  | 53 | 5 | 69 | E | 85 | U | 101 | e | 117 | u   |
| 6  | ACK | 22 | SYN | 38 | &  | 54 | 6 | 70 | F | 86 | V | 102 | f | 118 | v   |
| 7  | BEL | 23 | ETB | 39 | '  | 55 | 7 | 71 | G | 87 | W | 103 | g | 119 | w   |
| 8  | BS  | 24 | CAN | 40 | (  | 56 | 8 | 72 | H | 88 | X | 104 | h | 120 | x   |
| 9  | HT  | 25 | EM  | 41 | )  | 57 | 9 | 73 | I | 89 | Y | 105 | i | 121 | y   |
| 10 | LF  | 26 | SUB | 42 | *  | 58 | : | 74 | J | 90 | Z | 106 | j | 122 | z   |
| 11 | VT  | 27 | ESC | 43 | +  | 59 | ; | 75 | K | 91 | [ | 107 | k | 123 | {   |
| 12 | FF  | 28 | FS  | 44 | ,  | 60 | < | 76 | L | 92 | \ | 108 | l | 124 |     |
| 13 | CR  | 29 | GS  | 45 | -  | 61 | = | 77 | M | 93 | ] | 109 | m | 125 | }   |
| 14 | SO  | 30 | RS  | 46 | .  | 62 | > | 78 | N | 94 | ^ | 110 | n | 126 | ~   |
| 15 | SI  | 31 | US  | 47 | /  | 63 | ? | 79 | O | 95 | _ | 111 | o | 127 | DEL |

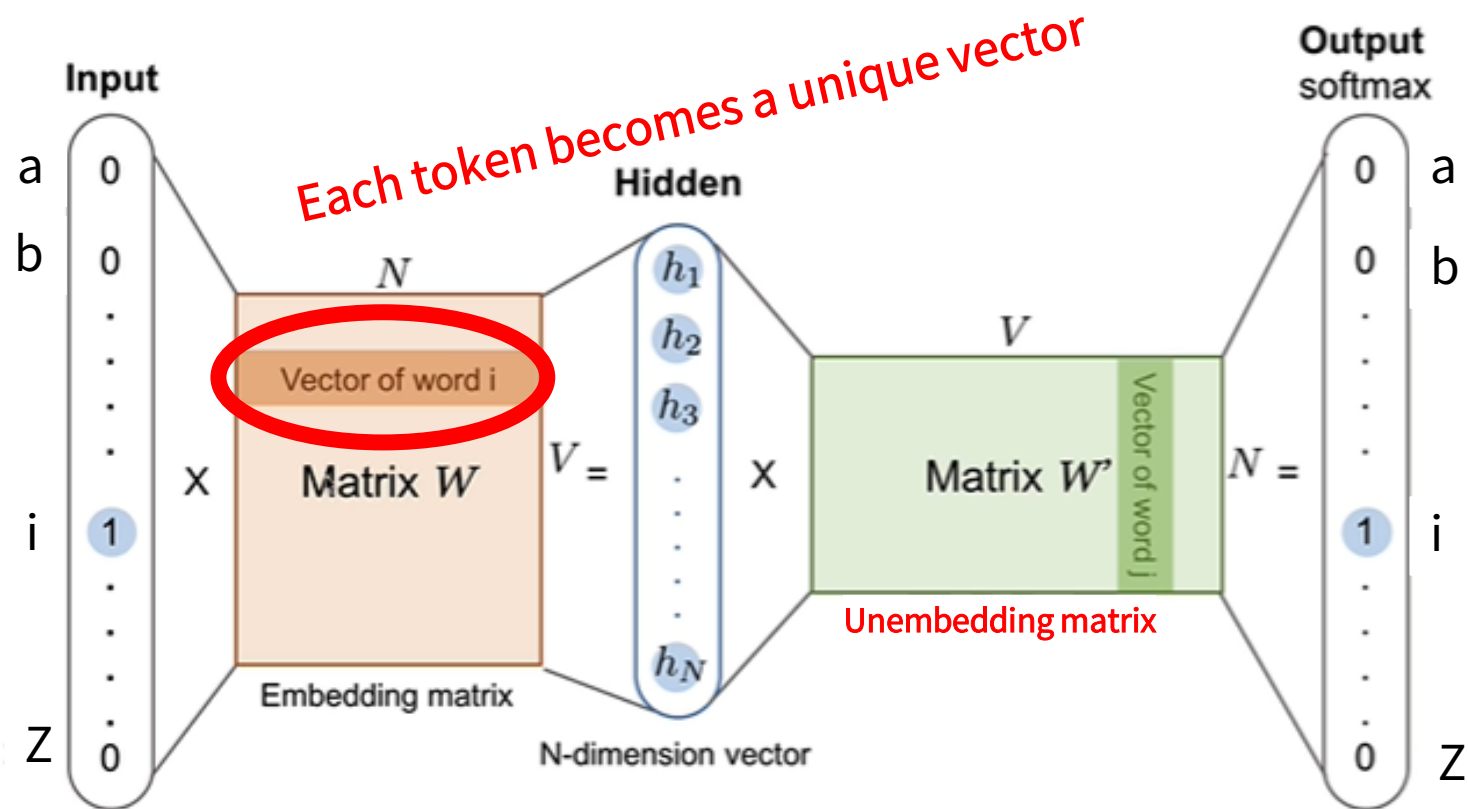
# What is Tokenization?

## ■ How computers process text?

| character | encoding | bits                                |
|-----------|----------|-------------------------------------|
| A         | UTF-8    | 01000001                            |
| A         | UTF-16   | 00000000 01000001                   |
| A         | UTF-32   | 00000000 00000000 00000000 01000001 |
| あ         | UTF-16   | 00110000 01000010                   |
| あ         | UTF-32   | 00000000 00000000 00110000 01000010 |

  
Vocabulary

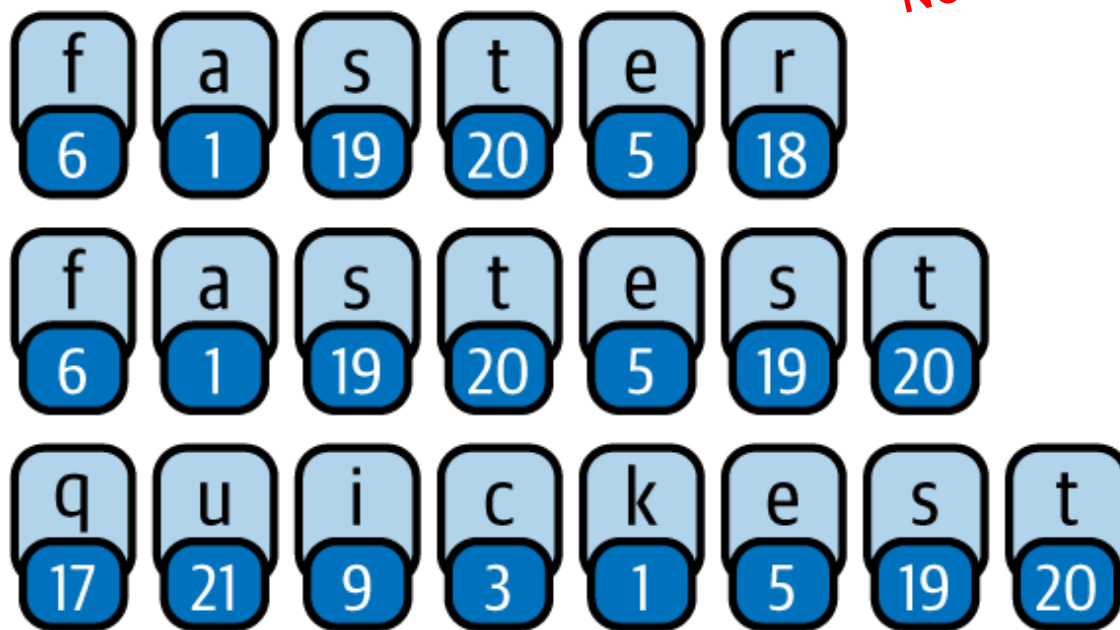
# What is Tokenization?



# What is Tokenization?

■ character  $\rightarrow$  token  $\rightarrow$  vector

## Character-based models

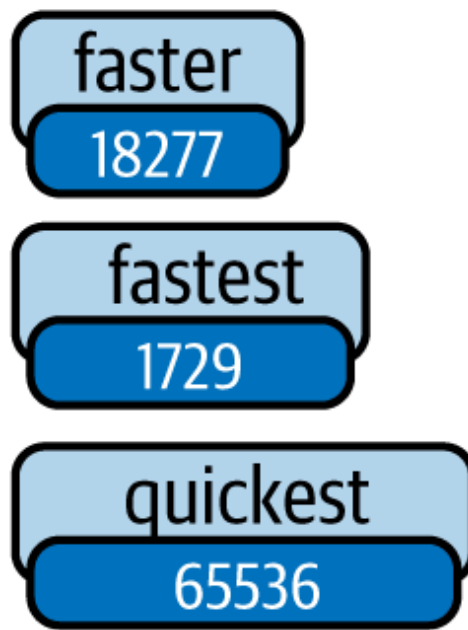


Not enough dimensions!

# What is Tokenization?

■ word  $\rightarrow$  token  $\rightarrow$  vector

## Word-based models



*Not enough memory!*

# What is Tokenization?

- We need to increase the Vocabulary
- But pay attention to the memory!

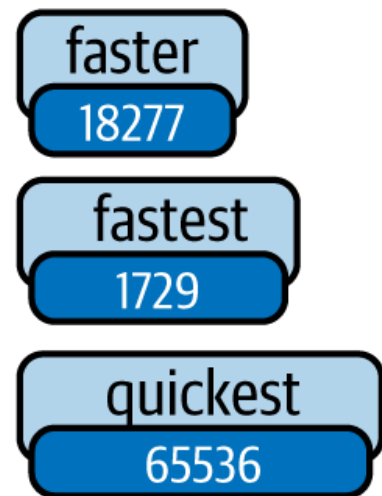
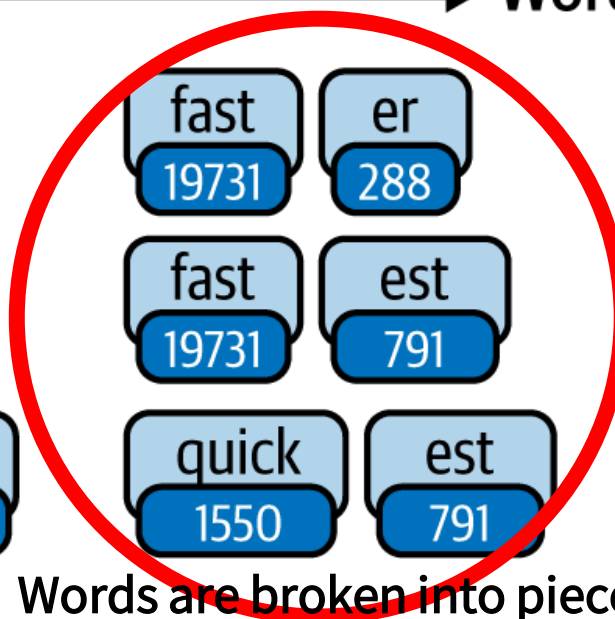
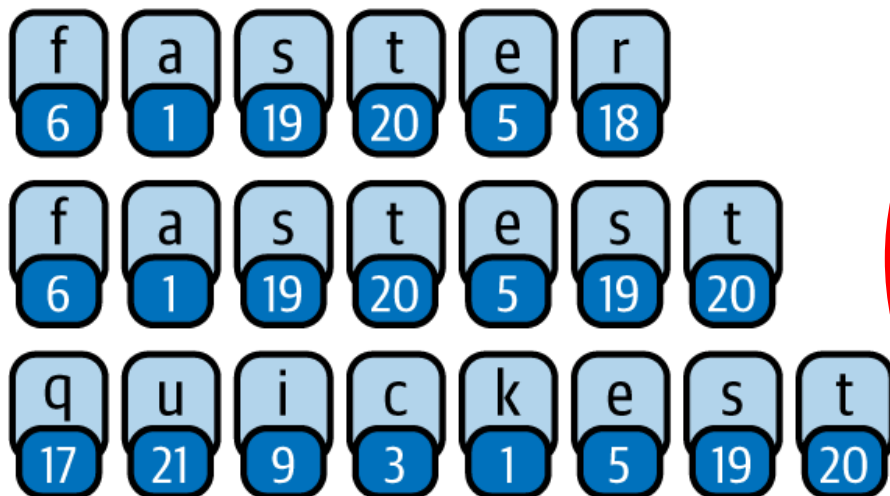
Not enough information

Character-based models

Vocabulary size

Too much information

Word-based models



Words are broken into pieces

# What is Tokenization?

## Tiktokenizer

|           |                                 |   |
|-----------|---------------------------------|---|
| System    | You are a helpful assistant     | × |
| User      | What do you like to eat?        | × |
| Assistant | I like to eat Camembert cheese. | × |

Token count  
36

```
<|im_start|>system<|im_sep|>You are a helpful assistant<|im_end|><|im_start|>user<|im_sep|>What do you like to eat?<|im_end|><|im_start|>assistant<|im_sep|>I like to eat Camembert cheese.<|im_end|><|im_start|>assistant<|im_sep|>
```

Camembert

```
200264, 17360, 200266, 3575, 553, 261, 10297, 29186,
200265, 200264, 1428, 200266, 4827, 621, 481, 1299, 3
16, 11237, 30, 200265, 200264, 173781, 200266, 40, 12
99, 316, 11237, 11349, 1634, 83, 23510, 13, 200265, 2
00264, 173781, 200266
```

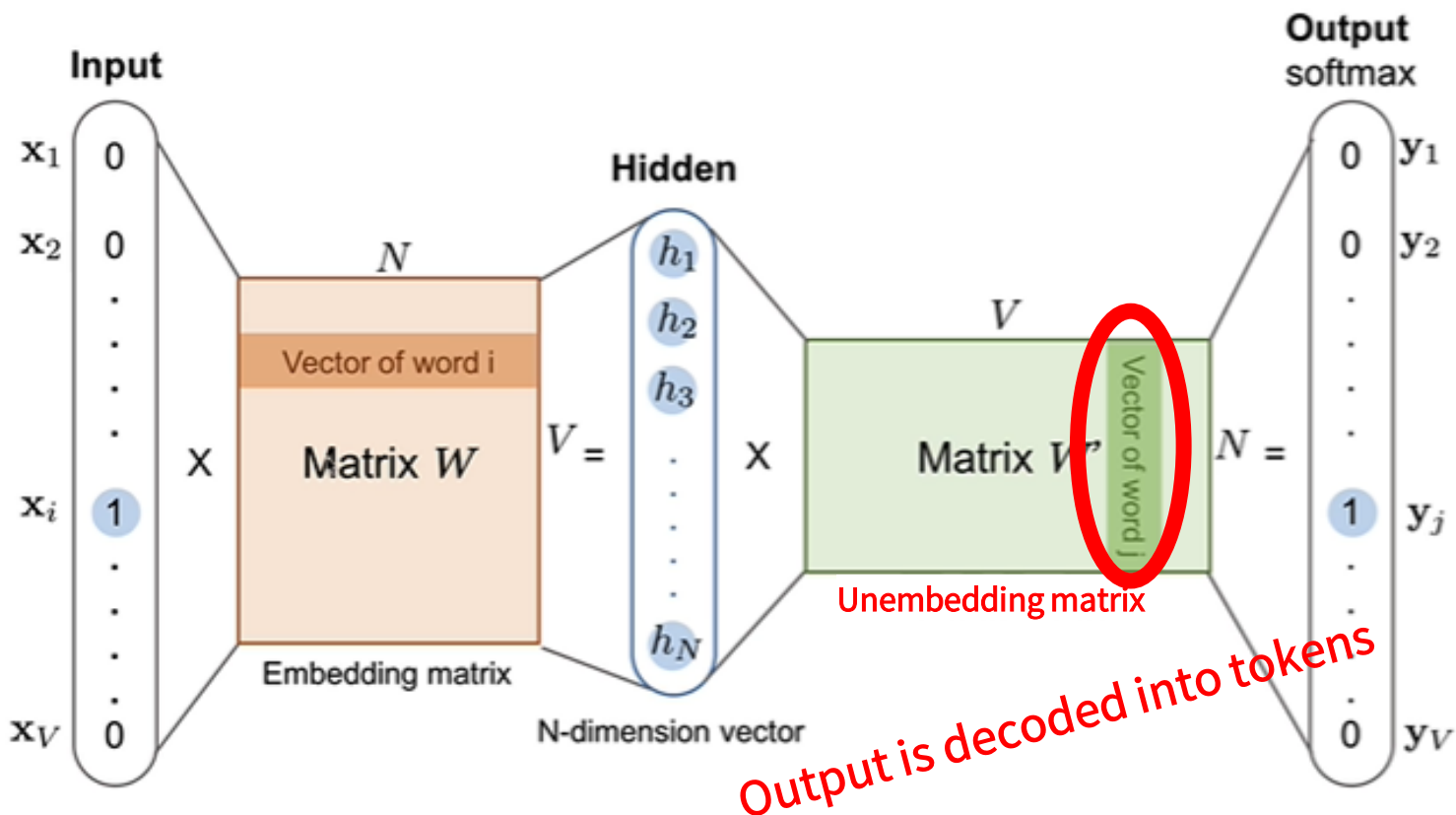


# Embedding\Unembedding Matrices

All Tokens  $\sim 50k$

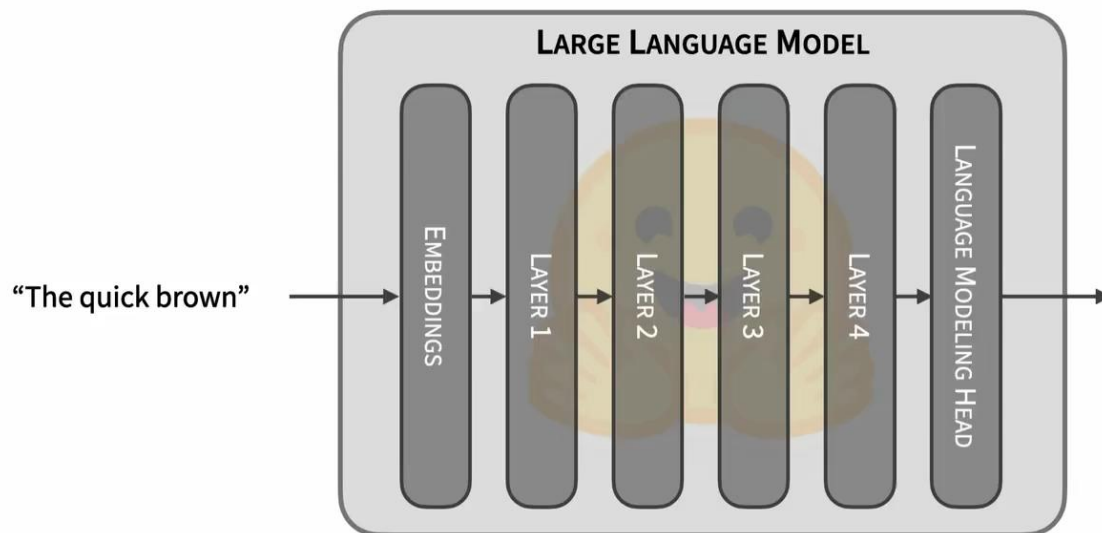
| aah  | aardvark | aardwolf | aargh | ab   | aback | abacterial | abacus | abalone | abandon | ... | zygoid | zygomat | zygomorphic | zygosis | zygote | zygotic | zyme | zymogen | zymosis | zzz  |
|------|----------|----------|-------|------|-------|------------|--------|---------|---------|-----|--------|---------|-------------|---------|--------|---------|------|---------|---------|------|
| +1.0 | +4.3     | +2.0     | +0.9  | -1.5 | +2.9  | -1.2       | +7.8   | +9.2    | -2.3    | ... | +0.6   | +1.3    | +8.4        | -8.5    | -8.2   | -9.5    | +6.6 | +5.5    | +7.3    | +9.5 |
| +5.9 | -0.8     | +5.6     | -7.6  | +2.8 | -7.1  | +8.8       | +0.4   | -1.7    | -4.7    | ... | -0.9   | +1.4    | -9.5        | +2.3    | +2.2   | +2.3    | +8.8 | +3.6    | -2.8    | -1.2 |
| +3.9 | -8.7     | +3.3     | +3.4  | -5.7 | -7.3  | -3.7       | -2.7   | +1.4    | -1.2    | ... | -7.9   | -5.8    | -6.7        | +3.0    | -4.9   | -0.7    | -5.1 | -6.8    | -7.7    | +3.1 |
| -7.2 | -6.0     | -2.6     | +6.4  | -8.0 | +6.7  | -8.0       | +9.4   | -0.6    | +9.4    | ... | +4.7   | -9.1    | -4.3        | -7.5    | -4.0   | -7.5    | -3.6 | -1.7    | -8.6    | +3.8 |
| +1.3 | -4.6     | +0.5     | -8.0  | +1.5 | +8.5  | -3.6       | +3.3   | -7.3    | +4.3    | ... | -6.3   | +1.7    | -9.5        | +6.5    | -9.8   | +3.5    | -4.6 | +4.7    | +9.2    | -5.0 |
| +1.5 | +1.8     | +1.4     | -5.5  | +9.0 | -1.0  | +6.9       | +3.9   | -4.0    | +6.2    | ... | +7.5   | +1.6    | +7.6        | +3.8    | +4.5   | +0.0    | +9.0 | +2.9    | -1.5    | +2.1 |
| -9.5 | -3.9     | +3.2     | -4.2  | +2.3 | -1.4  | -7.2       | -4.0   | +1.4    | +1.8    | ... | +3.0   | +3.0    | -1.4        | +7.9    | -2.6   | -1.3    | +7.8 | +6.1    | +4.0    | -7.9 |
| +8.3 | +4.2     | +9.9     | -6.9  | +7.3 | -6.7  | +2.3       | -7.4   | +6.9    | +6.1    | ... | -1.8   | -8.5    | +3.9        | -0.9    | +4.4   | +7.3    | +9.4 | +7.0    | -9.7    | -2.8 |
| ⋮    | ⋮        | ⋮        | ⋮     | ⋮    | ⋮     | ⋮          | ⋮      | ⋮       | ⋮       | ⋮   | ⋮      | ⋮       | ⋮           | ⋮       | ⋮      | ⋮       | ⋮    | ⋮       | ⋮       | ⋮    |
| -3.7 | -2.0     | -5.7     | -6.2  | +8.8 | +4.7  | -0.2       | -5.4   | -4.9    | -8.8    | ... | -3.7   | +3.9    | -2.4        | -6.3    | -9.4   | -8.6    | +3.6 | -0.9    | +0.7    | +7.9 |

# What is Tokenization?

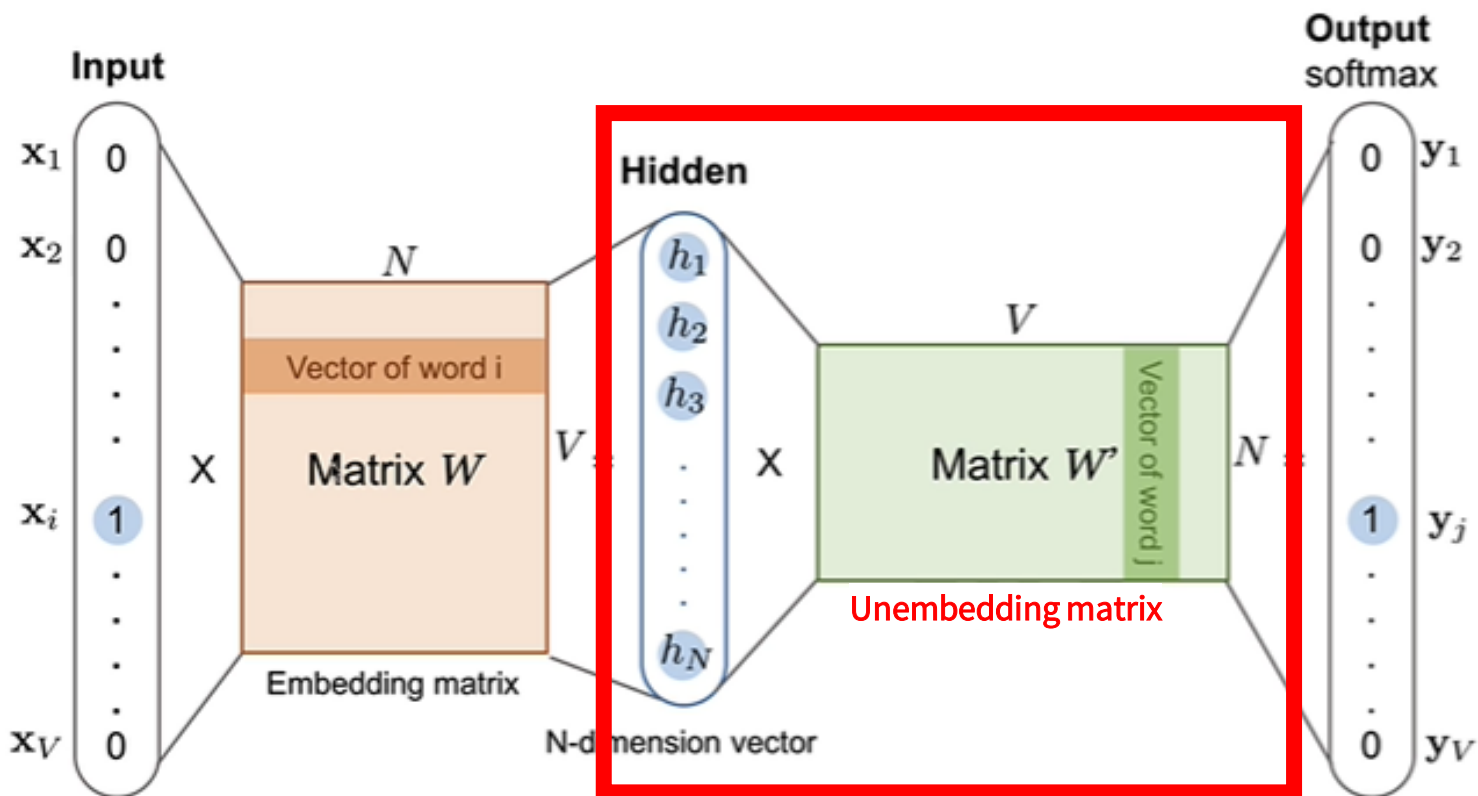


# Next token prediction

- The model predicts the probability for each possible token.



# What is Tokenization?



$$p(y_i|x) \propto e^{h(x)^T u(y_i)}$$

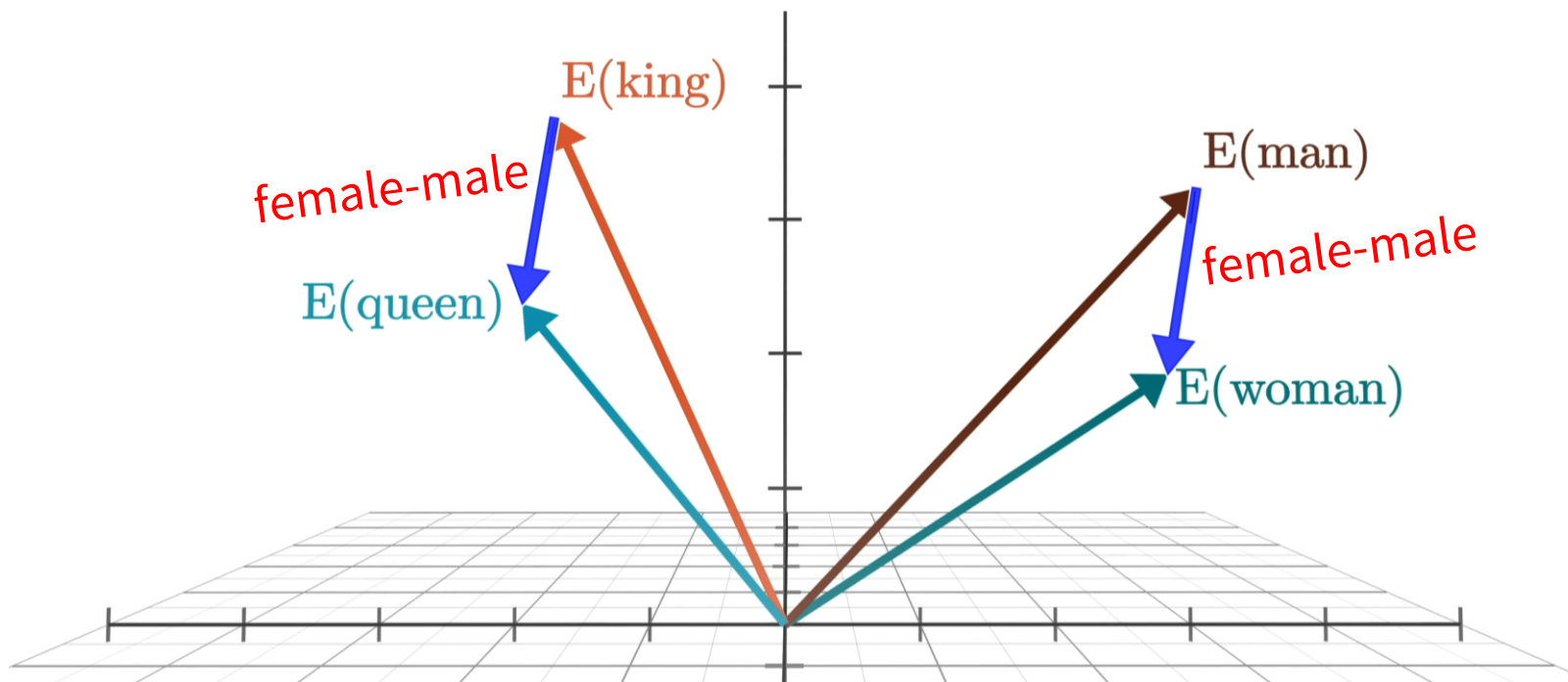
# What these vectors represent?

All Tokens  $\sim 50k$

| aah  | aardvark | aardwolf | aargh | ab   | aback | abacterial | abacus | abalone | abandon | ... | zygoid | zygomat | zygomorphic | zygosis | zygote | zygotic | zyme | zymogen | zymosis | zzz  |
|------|----------|----------|-------|------|-------|------------|--------|---------|---------|-----|--------|---------|-------------|---------|--------|---------|------|---------|---------|------|
| +1.0 | +4.3     | +2.0     | +0.9  | -1.5 | +2.9  | -1.2       | +7.8   | +9.2    | -2.3    | ... | +0.6   | +1.3    | +8.4        | -8.5    | -8.2   | -9.5    | +6.6 | +5.5    | +7.3    | +9.5 |
| +5.9 | -0.8     | +5.6     | -7.6  | +2.8 | -7.1  | +8.8       | +0.4   | -1.7    | -4.7    | ... | -0.9   | +1.4    | -9.5        | +2.3    | +2.2   | +2.3    | +8.8 | +3.6    | -2.8    | -1.2 |
| +3.9 | -8.7     | +3.3     | +3.4  | -5.7 | -7.3  | -3.7       | -2.7   | +1.4    | -1.2    | ... | -7.9   | -5.8    | -6.7        | +3.0    | -4.9   | -0.7    | -5.1 | -6.8    | -7.7    | +3.1 |
| -7.2 | -6.0     | -2.6     | +6.4  | -8.0 | +6.7  | -8.0       | +9.4   | -0.6    | +9.4    | ... | +4.7   | -9.1    | -4.3        | -7.5    | -4.0   | -7.5    | -3.6 | -1.7    | -8.6    | +3.8 |
| +1.3 | -4.6     | +0.5     | -8.0  | +1.5 | +8.5  | -3.6       | +3.3   | -7.3    | +4.3    | ... | -6.3   | +1.7    | -9.5        | +6.5    | -9.8   | +3.5    | -4.6 | +4.7    | +9.2    | -5.0 |
| +1.5 | +1.8     | +1.4     | -5.5  | +9.0 | -1.0  | +6.9       | +3.9   | -4.0    | +6.2    | ... | +7.5   | +1.6    | +7.6        | +3.8    | +4.5   | +0.0    | +9.0 | +2.9    | -1.5    | +2.1 |
| -9.5 | -3.9     | +3.2     | -4.2  | +2.3 | -1.4  | -7.2       | -4.0   | +1.4    | +1.8    | ... | +3.0   | +3.0    | -1.4        | +7.9    | -2.6   | -1.3    | +7.8 | +6.1    | +4.0    | -7.9 |
| +8.3 | +4.2     | +9.9     | -6.9  | +7.3 | -6.7  | +2.3       | -7.4   | +6.9    | +6.1    | ... | -1.8   | -8.5    | +3.9        | -0.9    | +4.4   | +7.3    | +9.4 | +7.0    | -9.7    | -2.8 |
| ⋮    | ⋮        | ⋮        | ⋮     | ⋮    | ⋮     | ⋮          | ⋮      | ⋮       | ⋮       | ⋮   | ⋮      | ⋮       | ⋮           | ⋮       | ⋮      | ⋮       | ⋮    | ⋮       | ⋮       | ⋮    |
| -3.7 | -2.0     | -5.7     | -6.2  | +8.8 | +4.7  | -0.2       | -5.4   | -4.9    | -8.8    | ... | -3.7   | +3.9    | -2.4        | -6.3    | -9.4   | -8.6    | +3.6 | -0.9    | +0.7    | +7.9 |

# Token Vectors have meaning

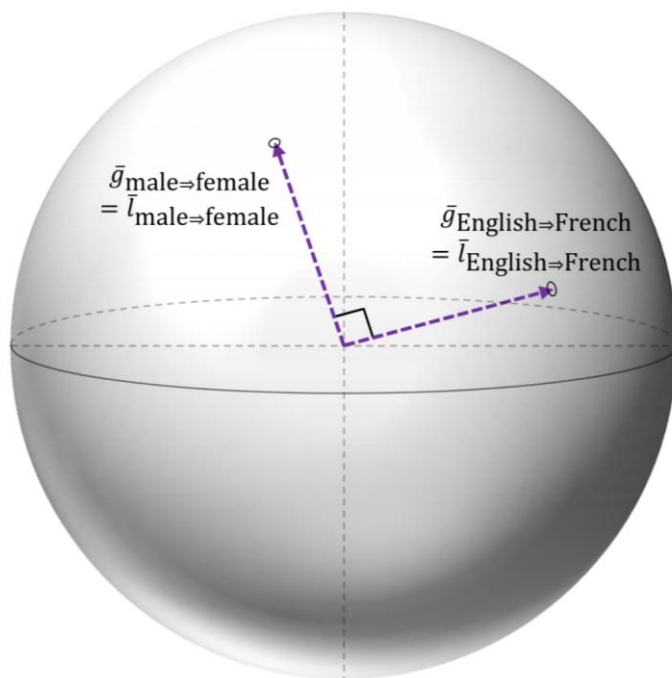
$$E(\text{queen}) - E(\text{king}) \approx E(\text{woman}) - E(\text{man})$$





# Linear Representation Hypothesis

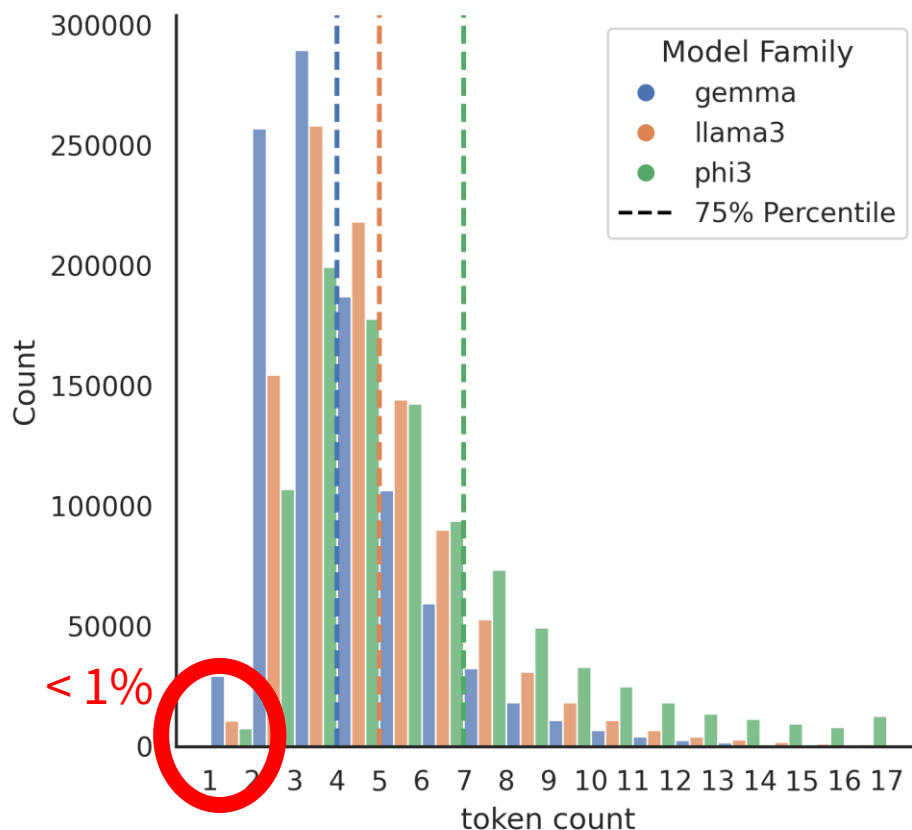
- Difference Vectors are Concept Subspaces.
- Unrelated concepts = orthogonal subspaces.





# Limitations of LRH

- Only works on words made of a single token



# Limitations of LRH

---

- Only works on 1-dim subspaces
- Only works on difference vectors



Then, let's just make high dimensional subspaces and difference subspaces

# Our Hypothesis

---

- Token Vectors  $\rightarrow$  Subspaces
- Difference Vectors  $\rightarrow$  Difference Subspaces
  - Token Vectors  $\rightarrow$  Subspaces
  - Difference Vectors  $\rightarrow$  Difference Subspaces

# Problem: We need to keep order.

| Meaning    | Word  |
|------------|-------|
| ad.n.01    | ad    |
| myth.n.01  | mit   |
| admit.v.01 | admit |
| half.n.02  | mitad |

“Admit” and “Mitad” would be the same subspace even though have different concepts

# How to make subspaces with order?

■ Flag = Sequence of subspaces

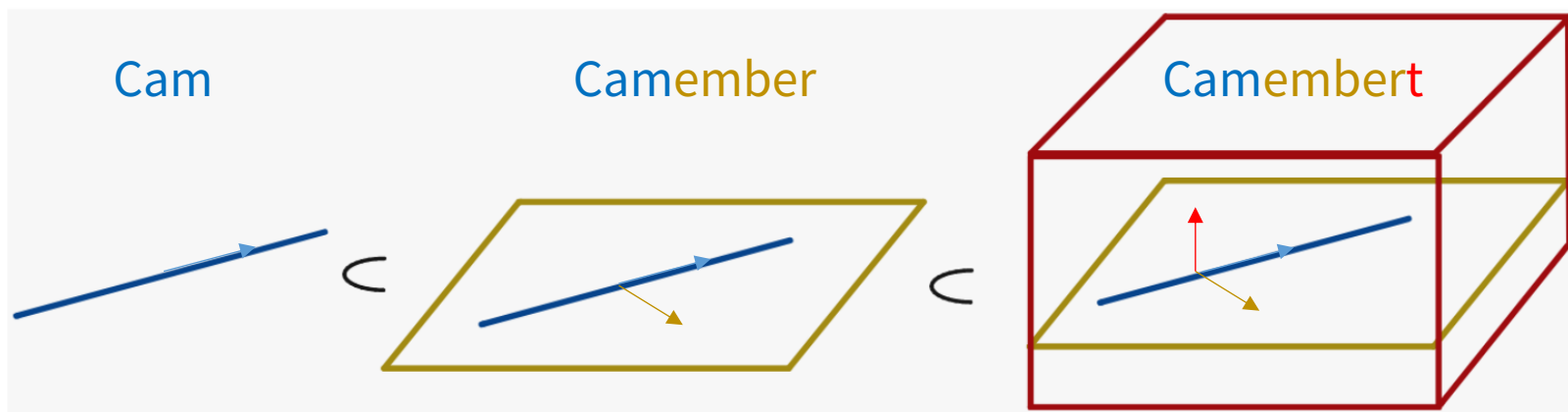
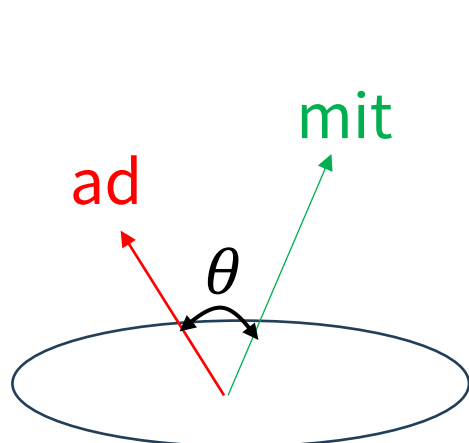


Figure 8: Illustration of a nested sequence of subspaces corresponding to a point on the flag manifold.

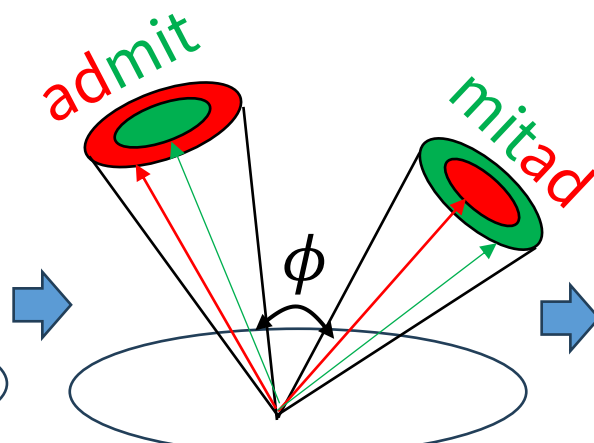
Tokens

Words

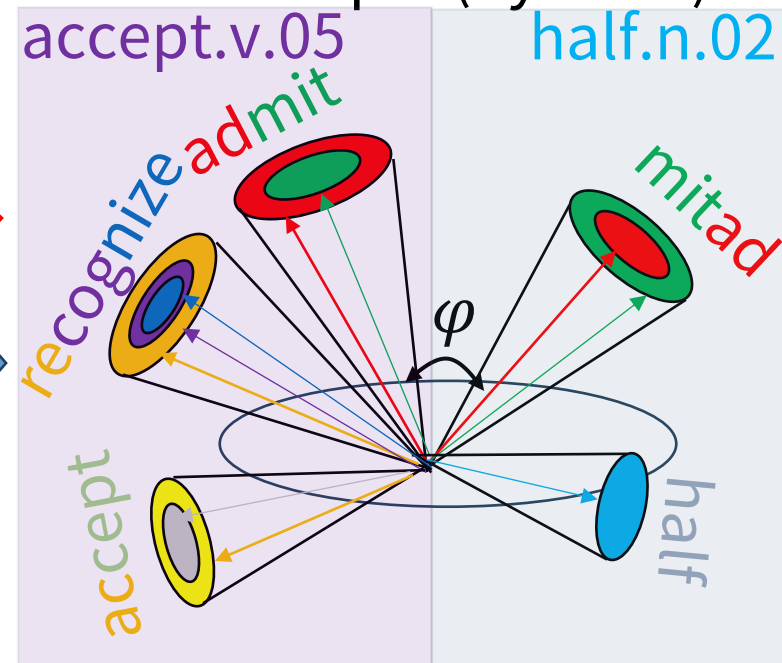
Concepts (Synsets)



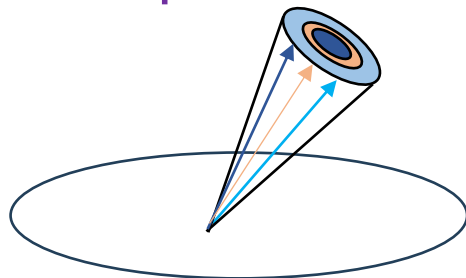
Subspaces



Flags

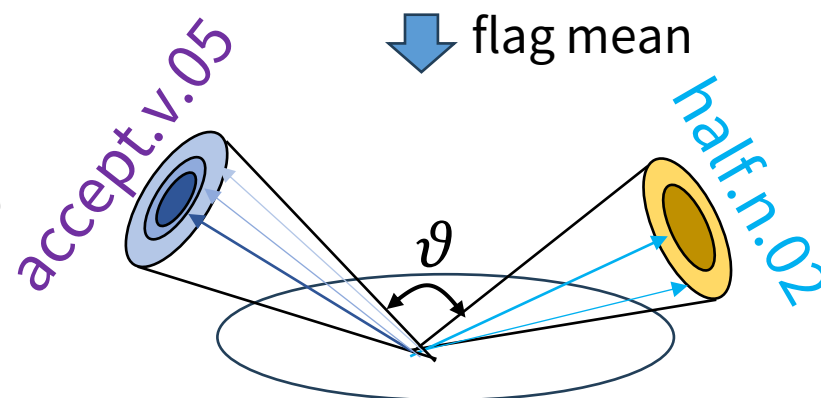


accept.v.05 - half.n.02



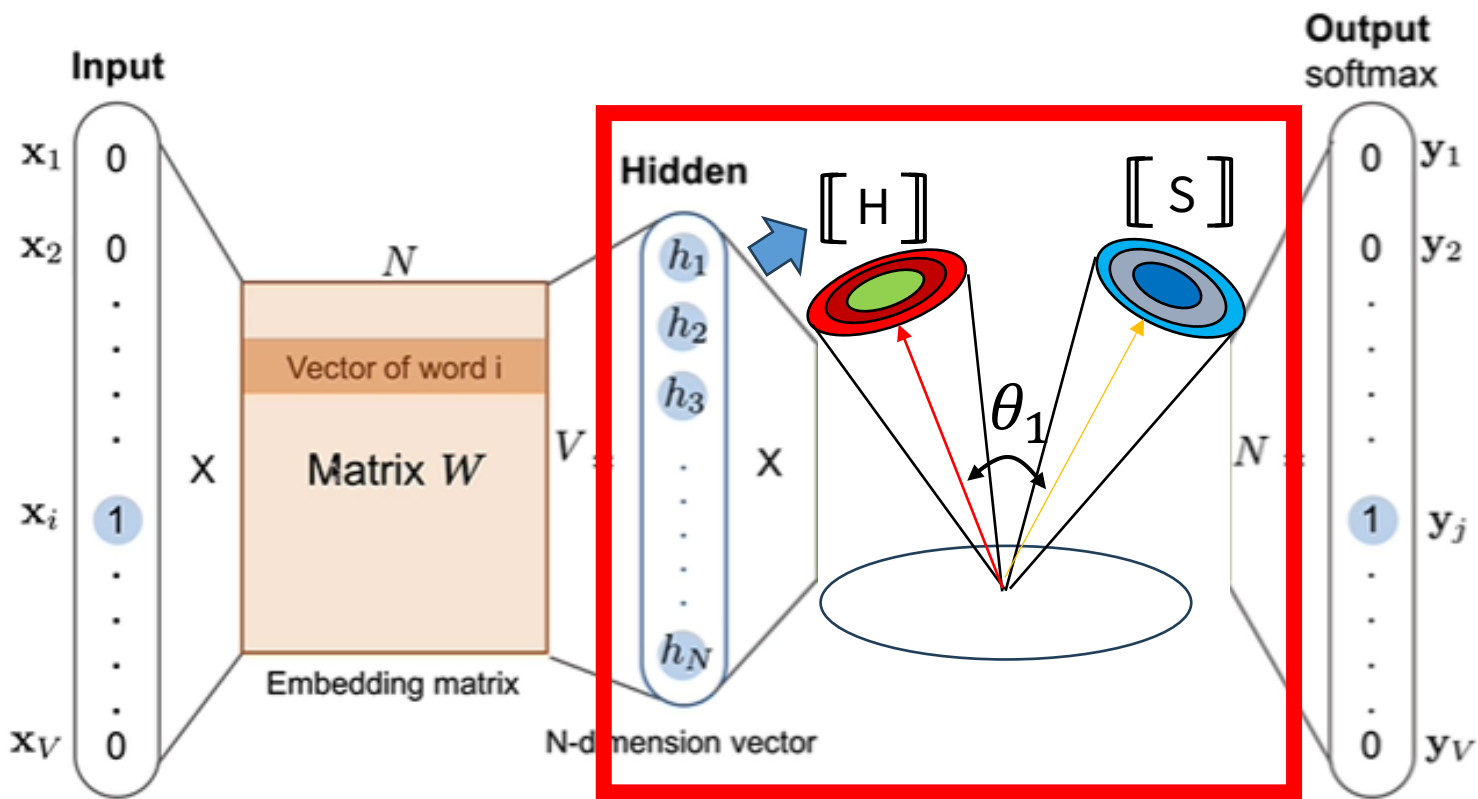
Higher Level  
Concept Flags

difference



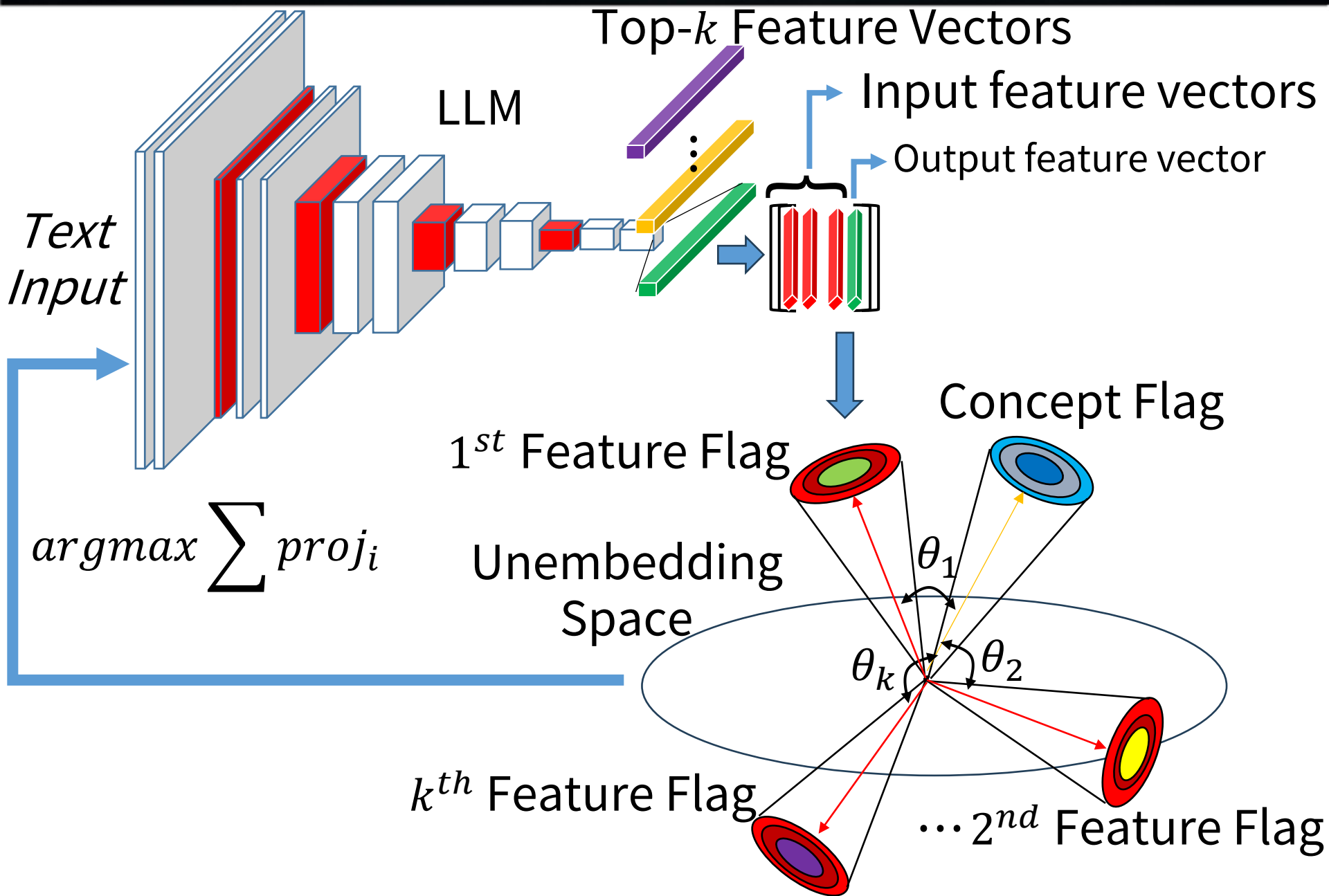
Concept Flags

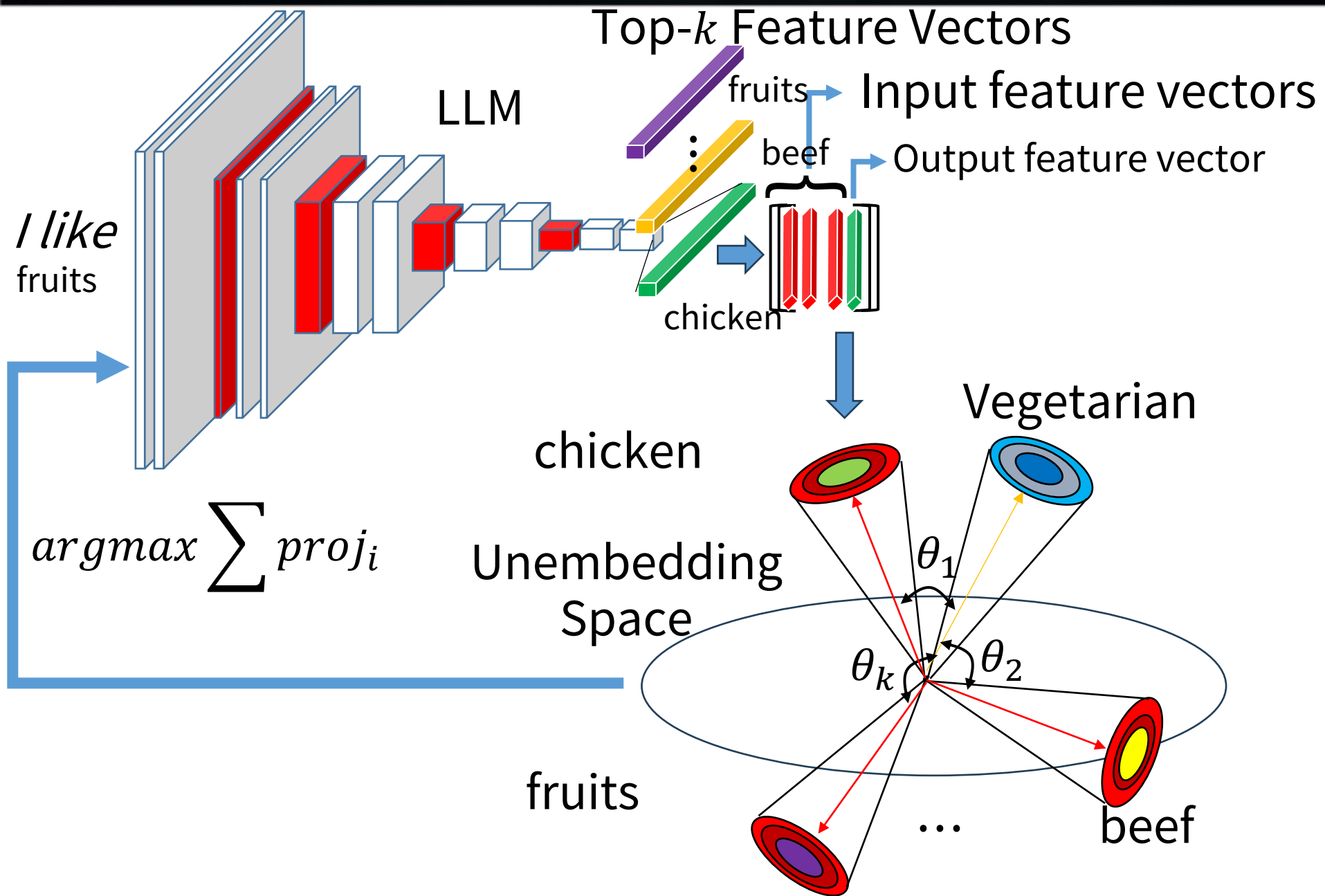
# Concept Probing



$$p(y_i|x) \propto e^{h(x)^T u(y_i)}$$







# Generation Results

 What men can be?

Men can be fathers, sons, brothers, and husbands. They can be role models and mentors, providing guidance...



no guidance

1. A husband. 2. A father. 3. A son. 4. A friend. 5. A boyfriend or partner...



woman.n.01 — male.n.01

Men as fathers and family providers, as well as caregivers, are essential for family well-being...



male.n.01 — woman.n.01

 What women can be?

Women can be mothers, wives, sisters, daughters, and friends.



no guidance

A human being of the female sex, typically having a vagina, breasts, and a womb.



woman.n.01 — male.n.01

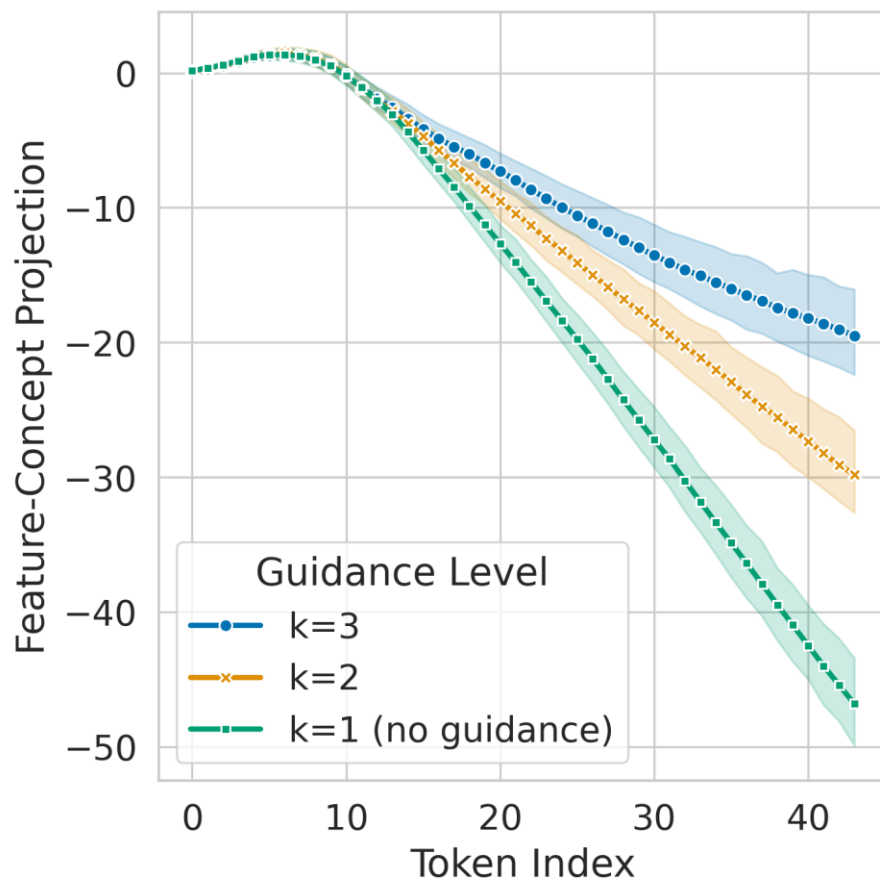
Women have been leaders in various capacities throughout human civilization. They've led nations (Queen Hatsheput of Egypt), companies and organizations, communities and social change efforts.



male.n.01 — woman.n.01

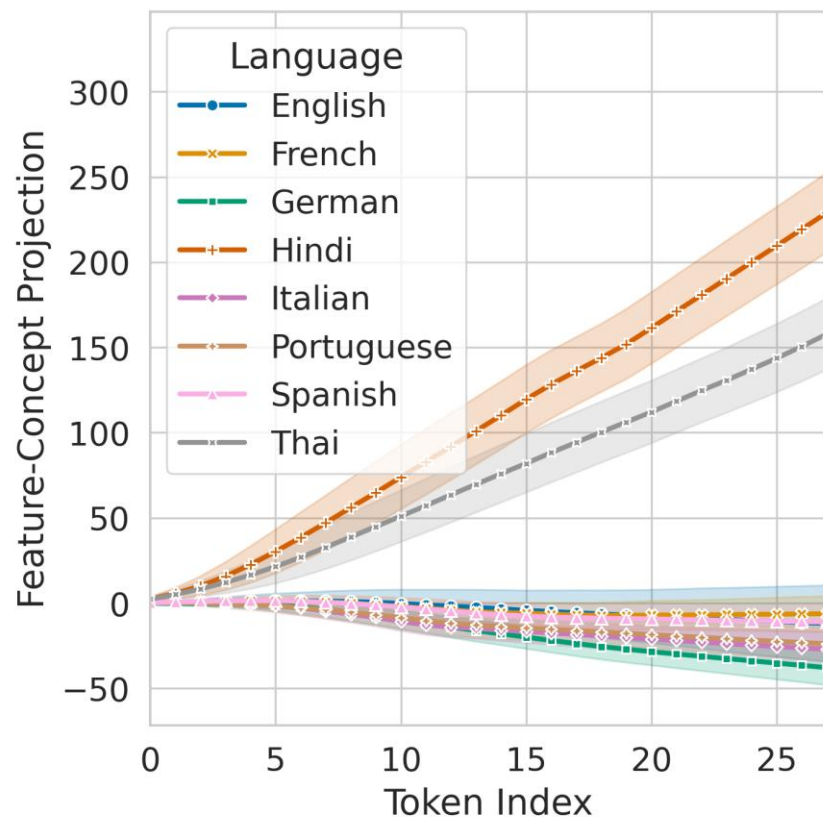
# Guidance Ablation

- We want to understand the effect of  $K$ .
- The higher the  $K$ , the more positive it becomes.
- **Conclusion:**  $K$  controls the guidance strength.



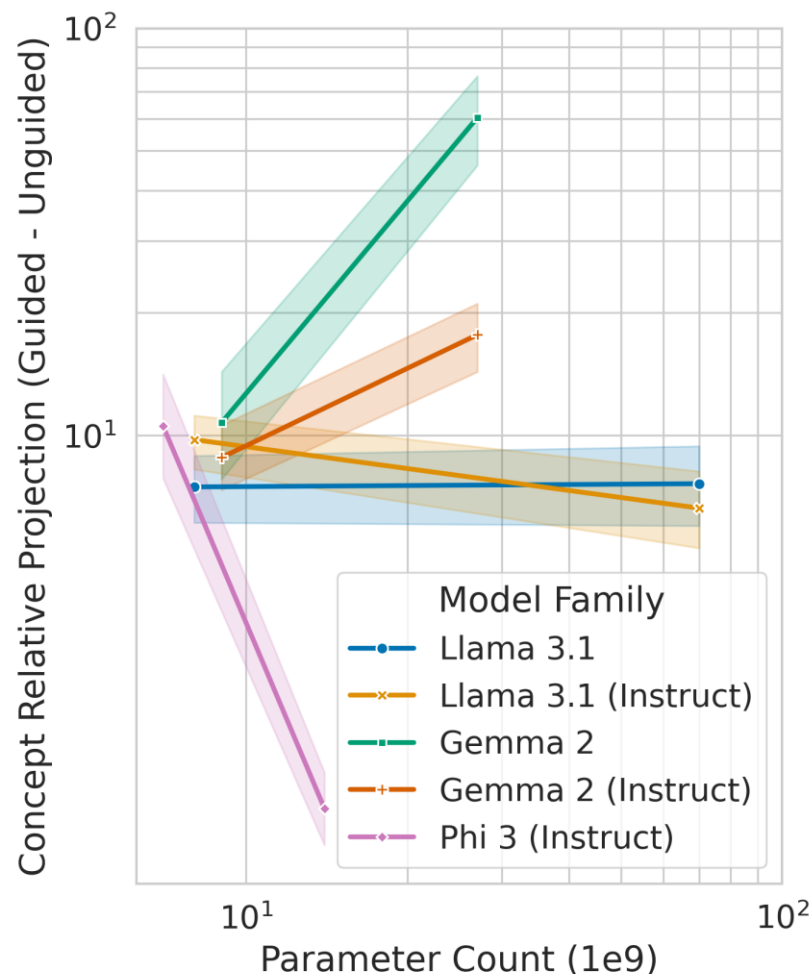
# Guidance Ablation

- We want to understand the effect on each language.
- **Conclusion:** Hindi and Thai are more susceptible to guidance.



# Guidance Ablation

- We want to understand the effect on each model.
- **Conclusion:** Gemma is the most susceptible, llama has average susceptibility and Phi the lowest.



---

# Thank you!